

Applied Security patches ↪ (update the code)

★ `sudo apt update`

(refreshed the packages (checks what updates are present)).

★ `sudo apt upgrade -y`

actually downloaded & installed those updates, including new kernel

★ `uname -r`

confirms that kernel was active

Configured the Firewall

★ Firewall - controls what network traffic is allowed in and out of your system.

Commands →

★ `sudo apt install ufw -y` (installed ufw)

★ `sudo apt apt ufw enable` (turned firewall on)

★ `sudo ufw default deny incoming`
(block all incoming connections by default)

★ `sudo ufw default allow outgoing`
(allow system to reach the internet normally)

★ `sudo ufw allow ssh`
(specifically allow ssh (remote login) through.)

★ `sudo ufw status verbose`
(shows final configuration)

Result

★ Default : deny (incoming), allow (outgoing)
22/tcp Allow IN Anywhere ← SSH access permitted

Disabled unused services

★ every running service is a potential attack surface - a way in for attacker if it has vulnerability.

- List all running services -
`sudo systemctl list-units --type=service --state=running`

(identified those that are not needed for the lab environment. (Modem Manager))

- Disable them -

`sudo systemctl stop <service>`

`sudo systemctl disable <service>`

`sudo systemctl mask <service>`